

Maintenance Work Order WO-2024-0342

Unit-2 Utilities Block | Mechanical Maintenance | Raised 2024-06-18

Work order	WO-2024-0342	Status	OPEN
Equipment	P-101 — Boiler Feed Water Pump P-101	Priority	P2 — Investigate within 7 days
Location	Unit-2 Utilities, Pump Bay B, Elev. +3.5 m	Work type	Investigation (no corrective work authorised)
Assigned to	Rajesh Kumar (EMP-1187)	Raised by	Anil Deshmukh (EMP-1533)
Trade	Rotating equipment	Planned hours	4.0
Related documents	INSP-2024-0315, ShiftLog 2024-06-14	Cost centre	UT2-MECH-01

1. Problem description

Vibration anomaly reported on P-101 following two high-vibration alarms on the night shift of 2024-06-14. The B Shift supervisor requested an investigation of the non-drive-end bearing condition, the lube oil supply and the discharge pressure trend. This work order authorises investigation and reporting only.

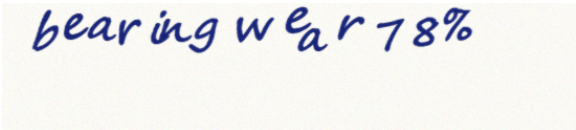
2. Findings

- NDE bearing wear measured at 78% of the OEM replacement limit of 85% (Sulzer CP 150-400 IOM, Section 7.4).
- Overall vibration 6.9 mm/s RMS at the NDE bearing housing (VT-1011), against a 7.1 mm/s alarm limit — 97% of the alarm setting.
- NDE bearing housing temperature 81 degC, up from 74 degC at the March 2024 inspection.
- Lube oil header pressure 1.8 bar; LP-101A discharge steady but 0.3 bar below commissioning.
- Shift log of 14 June 2024 records that the VT-1011 high-vibration alarm was bypassed twice during the night shift.
- Vibration spectrum shows 1x and 2x running-speed components consistent with the March 2024 inspection finding of a wear pattern similar to the 2022 failure.

3. Field measurements

Parameter	Measured	OEM limit	Percent of limit	Instrument / method
NDE bearing wear	78%	85%	92%	Shell-thickness comparison, Appendix D reference set
Overall vibration	6.9 mm/s RMS	7.1 mm/s	97%	VT-1011, portable collector
NDE bearing housing temp.	81 degC	95 degC	85%	Contact pyrometer
Lube oil header pressure	1.8 bar	1.4 bar min	129% of minimum	PI-1016

Technician's margin note (scanned from the field copy)



Handwritten annotation on the field copy of this work order, initialled RK. Transcribed value carried into Section 3 above.

4. Scheduling status

No bearing replacement work order has been raised and no bearing replacement is scheduled. This work order covers investigation only; a replacement work order against SOP-MECH-014 is recommended but not yet approved.

Follow-on work	Status	Reference
NDE / DE bearing replacement	NOT RAISED — not scheduled	SOP-MECH-014
Lube oil sample and analysis	Requested, not yet sampled	Lab request LR-2024-0611
VT-1011 alarm bypass removal	Open with instrumentation	Shift log 2024-06-14
LP-101A discharge trend review	Open with reliability	RCA-2022-0825

5. Recommendations

- Raise a bearing replacement work order against SOP-MECH-014 before wear reaches the 85% OEM limit; at the observed rate of approximately 7 percentage points per quarter the limit is reached in September 2024.
- Restore VT-1011 to service and remove the alarm bypass; brief night shift on the 2022 precedent.
- Draw a lube oil sample and compare iron content against the March 2024 result of 22 ppm.
- Inspect LP-101A discharge pressure trend — this was the root cause of the 2022 seizure.

6. History considered

The 2024-03-15 quarterly inspection (INSP-2024-0315) recorded 71% wear and noted a wear pattern similar to the 2022 failure. The 2022 event (INC-2022-0820) was an NDE bearing seizure that cost 14 hours of lost steam raising and Rs 25,00,000 (Rs 25 lakh). Root cause analysis RCA-2022-0825 attributed it to lubrication failure originating in LP-101A.

7. Sign-off

Executed by	Rajesh Kumar, Senior Technician — Rotating Equipment	Date	2024-06-18
Reviewed by	D. Krishnan, Maintenance Manager	Status at close	OPEN — investigation only

This work order remains OPEN. No corrective maintenance has been authorised against it. Any bearing replacement requires a separate work order raised against SOP-MECH-014 Rev 3.